

• PRODUCT REQUIREMENTS DOCUMENT

DEVCON+ PM Dashboard

A purpose-built project management platform for DEVCON+ Philippines — designed ground-up to match the team's exact workflow.

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Executive Summary

DEVCON+ PM is a **self-hosted, custom-built project management dashboard** purpose-built for the internal operations of DEVCON+ Philippines. Rather than bending a third-party tool (Jira, Notion, Linear) to fit the team's workflow, DEVCON+ PM is designed ground-up to match exactly how this team plans, tracks, communicates, and ships.

The product is a **full-stack web application** built on Next.js 14 App Router + Supabase, deployed on Vercel, accessible from any browser — no installation required. It replaces ad-hoc spreadsheet tracking and scattered communication with a single, role-aware, real-time source of truth.

Design Philosophy: Every feature decision is shaped by one question — does this match how the DEVCON+ team actually works? Nothing is added for feature parity with other tools. Everything exists because the team needs it.

Core Capabilities at a Glance

- **Open Kanban Board** — accessible without login, real-time sync via Supabase Realtime
- **Role-aware Access Control** — three tiers: Admin, Contributor, Guest
- **Proactive Email Alerts** — admin notified of all non-admin board activity within 60s
- **QA & Bug Traceability** — QA tests and bugs linked directly to tasks
- **Meeting, Milestone & Announcements** — integrated team coordination modules
- **Essentials Knowledge Base** — project-scoped, supports credentials, links, files, code

Problem Statement

Core Pain Points

PAIN POINT	CURRENT STATE	IMPACT
Task visibility	Scattered across chats and spreadsheets	Work gets missed or duplicated
Status tracking	Manual updates, no enforcement	Stale data, no accountability

PAIN POINT	CURRENT STATE	IMPACT
QA traceability	QA results not linked to tasks or bugs	Fixes ship without verified test coverage
Bug management	Informal, no severity or ownership	Critical issues deprioritized
Contributor access	No formal onboarding or offboarding	Security and coordination risk
Meeting coordination	Separate tools, no project context	Meeting action items lost
Announcements	Chat messages buried in history	Team misses critical updates
Admin overhead	No central management UI	Manual effort to add/remove people

Why Not an Off-the-Shelf Tool?

Existing PM tools either over-engineer the workflow (Jira), require costly subscriptions (Linear), or lack the specific combination of:

- Open board access for contributors without login friction
- Admin-only management controls for team membership
- Native email and Telegram notifications tied to project events
- A UI shaped around DEVCON+'s exact task lifecycle

Goals & Success Metrics

Primary Goals

- Give the admin full visibility and control over all team activity, contributors, and project state
- Allow contributors to manage their work on the board without login friction
- Ensure every significant board action is tracked and the admin is proactively notified of all non-admin activity
- Replace all external PM tooling for the DEVCON+ team

Success Metrics

100%

Task tracking adoption

All active tasks tracked on the board

< 2 min

Contributor onboarding time

Admin adds → welcome email sent

60s

Alert delivery SLA

Email alert for any non-admin board action

> 80%

QA test coverage

Shipped features with a linked QA test

Users & Roles

4.1 Role Definitions



Admin

rizzperocho@gmail.com

- Full read/write access across all modules
- Sole access to Contributors management
- Real-time in-app activity feed (bell)



Contributor

Added by admin to contributors table

- Access to all pages except Contributors management
- Full edit rights on board tasks and groups



Guest

Unauthenticated visitor

- Full read/write access to the board (open board model)
- No access to Contributors management (server-side redirect)

- Receives all non-admin email alerts
- Admin bypass if contributor record missing
- Login via Google OAuth or email/password

- Full edit rights on QA, Bugs, Announcements, Meetings, Milestones, Essentials
- Receives email when assigned to a task
- Welcome email sent automatically on addition
- Redirected to /dashboard on login

- Notification bell hidden
- Sidebar shows "Viewing as Guest" + "Sign in as Admin" link
- All guest actions logged + admin email alerts triggered
- Identified as "Guest" in activity notifications

Design Rationale (Open Board Model): The open board model removes login friction for contributors/collaborators while keeping the sensitive Contributors page locked to the admin. All non-admin activity is monitored, so the admin maintains full awareness without restricting participation.

Access Control Matrix

FEATURE	ADMIN	CONTRIBUTOR	GUEST
PM Board — view	✓	✓	✓
PM Board — edit tasks/groups	✓	✓	✓
QA Tracker	✓	✓	✓
Bug Tracker	✓	✓	✓
Announcements	✓	✓	✓
Meetings	✓	✓	✓
Milestones	✓	✓	✓
Essentials	✓	✓	✓
Contributors page	✓	✗ redirect	✗ redirect
Notification bell	✓ All activity	Assignments only	✗ Hidden
Activity email alerts	✓ Receives	✗	✗
Supabase admin bypass	✓	✗	✗

Feature Requirements

6.1 Authentication & Session Management

FR-AUTH-1 Email/Password Login

Users authenticate with email + password via Supabase Auth. Invalid credentials return an inline error. On success, the user's contributor record is fetched from the `contributors` table and stored in Zustand (`useAuthStore`).

FR-AUTH-2 Google OAuth

Users can sign in via Google. The OAuth callback route (`/auth/callback`) exchanges the code for a session, verifies the contributor record, and redirects to `/dashboard`. Admin email bypasses the contributor check.

FR-AUTH-3 Admin Bypass

If `rizzperoch@gmail.com` is authenticated but has no contributor record in the DB, a synthetic contributor object is created in the Zustand store (`id: "admin"`) so the UI functions normally. This prevents admin logout.

FR-AUTH-4 Access Denied

Any authenticated user whose email is NOT in the `contributors` table AND is NOT the admin email is immediately signed out and redirected to `/access-denied`.

FR-AUTH-5 Open Board (Guest Access)

Unauthenticated visitors are not redirected to `/login`. The middleware only redirects on: authenticated user visiting `/login` → send to `/dashboard`. All other routes are accessible without auth.

FR-AUTH-6 Session Hydration

On mount, `AuthProvider` calls `supabase.auth.getUser()`. If a session exists, it fetches the contributor record. If no session: sets `contributor = null` (guest mode). The `SIGNED_OUT` auth event triggers a redirect to `/login` — but `INITIAL_SESSION` with null (guest visits) does not.

6.2 PM Board (Kanban)

FR-BOARD-1 Projects

Users can create projects (name + optional description). Projects appear in a left sidebar. Clicking a project loads its groups and tasks. First project is auto-selected on load.

FR-BOARD-2 Groups

Projects contain Groups (analogous to sprints or feature areas). Groups contain Tasks, rendered as table rows.

- **Add group:** inline form at the bottom of the board
- **Rename group:** click group name → inline edit → blur/Enter to save
- **Delete group:** trash icon (hover) → confirmation modal (warns all tasks will be deleted)
- **Reorder groups:** drag handle (⋮, hover-visible) using `@dnd-kit`
- **Collapse/expand:** chevron icon (animated rotation) — toggles task table visibility

FR-BOARD-3 Group Color Coding

Each group has a manually assigned accent color from a 12-color palette: Blue, Violet, Indigo, Teal, Emerald, Lime, Amber, Orange, Red, Rose, Pink, Cyan.

- A colored circle button in the group header opens a 4×3 swatch grid
- Selected color indicated with a white checkmark overlay
- Color persists in localStorage keyed by `devcon-group-color-{groupId}`
- On first render (no localStorage entry), a default color is assigned from the rotating palette based on group index
- Accent color applies to: left border (4px), tinted header background (~7% opacity), group name text

FR-BOARD-4 Tasks

Each task is a row in the group table with these columns:

COLUMN	BEHAVIOR
Task (Title)	Inline edit on click, blur to save
Assignee	Dropdown showing all contributors (portal-rendered, no scroll clipping)
Status	Color-coded badge dropdown (portal-rendered)
Timeline	Date range picker (start → end)

COLUMN	BEHAVIOR
Due Date	Single date picker
Attachment	Upload files to Supabase Storage, view/delete
PR Link	Inline URL input

Task actions (hover-visible): drag to reorder (within group), delete (confirmation modal), expand description (textarea below the row).

FR-BOARD-5 Task Status System

Six statuses with distinct colors:

Not Started

#9ca3af

In Progress

#3b82f6

Done

#22c55e

Help

#f59e0b

I am Stuck

#ef4444

For Improvements

#a855f7

Each task row has a 3px colored left-border stripe matching its status for at-a-glance scanning.

FR-BOARD-6 Drag & Drop

Tasks reorder within a group via vertical DnD (`@dnd-kit` , `PointerSensor`, 8px activation distance). Groups reorder within a project via vertical DnD. Position values are written back to Supabase on drop.

FR-BOARD-7 Portal Dropdowns

Assignee and Status dropdowns render via `ReactDOM.createPortal` at `document.body` with `position: fixed` , positioned using `getBoundingClientRect()` . This prevents clipping by the `overflow-x-auto` table wrapper. Dropdowns auto-flip upward when less than ~260–280px of space exists below the trigger.

FR-BOARD-8 Real-time Board Sync

The board subscribes to Supabase Realtime `postgres_changes` for the selected project's tasks table. INSERT/UPDATE/DELETE events are applied to local state without a full reload. Multiple users editing the same board see changes in real-time.

6.3 Activity Logging & Notifications

FR-NOTIF-1 Activity API

All CRUD operations on the board call `POST /api/activity` (fire-and-forget). Payload: `{ action, entity, entityTitle, actorName, actorEmail, page }`.

Actions logged: task created, updated (status/assignee/title/due date changes), moved (group change), deleted; group created, renamed, deleted; project created. Position-only reorders are not logged (too noisy).

FR-NOTIF-2 Admin Email Alerts

`/api/activity` checks `isAdmin(actorEmail)`. For non-admin actors (contributors or guests): sends a styled HTML email to `rizzperoch@gmail.com` via Resend. Email contains: actor name, action, entity title, page, and timestamp in PHT (Asia/Manila). Color-coded left border: green=created, red=deleted, blue=all others. Admin's own actions do not trigger an email.

FR-NOTIF-3 In-App Notification Bell (Admin)

Shown only to the admin. Subscribes to Supabase Realtime on tasks (INSERT/UPDATE/DELETE, all projects) and groups (INSERT/DELETE). UPDATE events only surface meaningful field changes (status, assignee, title, group_id). Displays an "Activity Feed" dropdown with up to 30 entries. Shows unseen count badge (red). "Clear all" button. Clicking an activity entry navigates to `/dashboard`.

FR-NOTIF-4 In-App Notification Bell (Contributor)

Shows assignment notifications only. Subscribes to tasks Realtime filtered by `assignee_id = contributor.id`. INSERT and UPDATE events on that filter trigger a notification. Displays "Notifications" label.

FR-NOTIF-5 Task Assignment Email

When a task's `assignee_id` changes to a non-null value, an email is sent to the new assignee via `POST /api/notify` (type: `task_assigned`). This is in addition to the activity alert sent to the admin.

6.4 Contributors Management

FR-CONTRIB-1 Access Guard

The `/contributors` route performs a server-side check: `user.email !== ADMIN_EMAIL` → `redirect("/dashboard")`. Enforced regardless of Supabase session state.

FR-CONTRIB-2

Contributor List

Table showing: avatar (initials), full name, email, role badge, Telegram handle, join date. Hover reveals Edit and Remove actions.

FR-CONTRIB-3

Add Contributor

Modal form: full name, email, role (dropdown), Telegram username. On submit: inserts row in `contributors` table, sends welcome email via Resend (type: `welcome_contributor`).

FR-CONTRIB-4

Edit Contributor

Same form fields as add. Updates the DB row in place.

FR-CONTRIB-5

Remove Contributor

Soft delete: sets `deleted_at` timestamp. Contributor immediately loses access (login check filters `deleted_at IS NULL`). Browser confirm dialog before action.

FR-CONTRIB-6

Roles Management

Side panel for managing roles (name, description, color). Roles are assigned to contributors and displayed as colored badges throughout the UI (sidebar, task assignee cell).

| 6.5 QA Tracker

Test case management per project. Each QA test has: title, description, category, status (Pass/Fail/Blocked/Not Run), assignee, bug report text, linked bug ID. Supports creating, editing, and bulk status updates. Status changes trigger email notifications to the assignee.

| 6.6 Bug Tracker

Structured bug reporting with the following fields:

- **Core fields:** title, description, steps to reproduce, expected vs actual behavior
- **Classification:** severity (Critical/High/Medium/Low), status (Open/In Progress/Resolved/Closed/Cannot Reproduce)
- **Assignment:** reporter, assignee — bug assignment triggers email notification
- **Context:** environment, browser/device, screenshot URLs, PR link
- **Links:** linked task, linked QA test

Activity log per bug: tracks field changes (who changed what, old → new value, timestamp).

| 6.7 Announcements

Create and broadcast announcements to the team. Each announcement: title, body (supports line breaks), created by, sent_at timestamp. Announcements can be sent via email broadcast to all contributors via Resend.

| 6.8 Meetings

Schedule and manage team meetings with:

- **Details:** title, type (Standup/Audit/Other), description, date, start/end time, timezone
- **Recurrence:** None/Daily/Weekly/Biweekly/Monthly with recurrence end date
- **Integration:** Google Meet link, Google Calendar event ID
- **Reminders:** reminder offset (minutes before) via Edge Function

Attendee RSVPs tracked per meeting. Email notifications: meeting confirmation on schedule, cancellation notice, and reminder N minutes before start. Calendar grid view + detail modal.

| 6.9 Milestones

Track high-level project goals. Each milestone: title, description, target date, status (Not Started/In Progress/At Risk/Achieved/Missed), achieved_at, announced flag, linked project.

Progress logging: contributors log progress notes, percentage completion, and blockers. History modal shows the full progress timeline. When a milestone is marked "Achieved," an email broadcast is sent to all

6.10 Essentials (Knowledge Base)

A project-scoped knowledge base. Organized as Sections (groupings with title, icon, description) containing Entries.

- **Entry types:** text, link, code, file, email, credential
- **Sensitive entries:** can be marked `is_sensitive` (masked by default)
- **Files:** stored in Supabase Storage with signed URLs
- Supports CRUD on both sections and entries, with drag-reorder by position

Technical Architecture

Technology Stack

LAYER	TECHNOLOGY
Frontend	Next.js 14 (App Router, React Server Components + Client Components)
Auth	Supabase Auth (email/password + Google OAuth)
Database	Supabase (PostgreSQL) with Row Level Security
Storage	Supabase Storage (task attachments, essential files)
Real-time	Supabase Realtime (postgres_changes)
State	Zustand (useAuthStore)
Email	Resend (transactional + broadcast)
Deployment	Vercel
Styling	Tailwind CSS
Drag & Drop	@dnd-kit/core + @dnd-kit/sortable

Key Architectural Decisions

ARCH-1

Server vs Client Components

Data-fetching pages (Dashboard, QA, Contributors, etc.) are async Server Components that hydrate the client with initial data via props, avoiding client-side loading spinners on first render. Interactivity (board mutations, modals, dropdowns) is in `"use client"` components.

ARCH-2

Auth Guard Layering (3 Layers)

- **Middleware:** session cookie refresh + login → dashboard redirect
- **AuthProvider:** client-side session hydration, admin bypass, access-denied flow
- **Page-level (server):** Contributors page verifies `user.email === ADMIN_EMAIL`, redirects otherwise

ARCH-3 Permissions Module

`lib/permissions.ts` — Single source of truth for `ADMIN_EMAIL` constant and `isAdmin()` helper, imported across middleware, server components, client components, and API routes.

ARCH-4 Portal Dropdowns

All floating UI elements (status picker, assignee picker, color picker) that live inside `overflow-x-auto` containers use `ReactDOM.createPortal` + `position: fixed` to escape overflow clipping. Auto-flip logic prevents off-screen rendering.

ARCH-5 Activity System

Fire-and-forget `POST /api/activity` calls from client components. API route is the single gating point for admin email alerts — never blocks the UI on email delivery failure (always returns `{ ok: true }`).

ARCH-6 Group Colors (localStorage)

Stored in localStorage (key: `devcon-group-color-{groupId}`) to avoid requiring a DB schema change. Hydrated client-side after mount to prevent SSR hydration mismatch.

Database Schema

```
roles → id, name, description, color, created_at contributors → id, email, full_name,
role_id, telegram_username, deleted_at, created_at projects → id, name, description,
created_by, created_at groups → id, project_id, name, position, created_at tasks → id,
group_id, project_id, title, description, assignee_id, status, timeline_start,
timeline_end, due_date, pr_link, position, created_at, updated_at task_attachments → id,
task_id, file_name, file_url, uploaded_by, uploaded_at announcements → id, title, body,
created_by, sent_at, created_at qa_tests → id, project_id, title, description, category,
status, assigned_to, bug_report, bug_id, created_at, updated_at bugs → id, project_id,
title, description, steps_to_reproduce, expected_behavior, actual_behavior, severity,
status, reported_by, assigned_to, qa_test_id, task_id, pr_link, environment,
browser_device, screenshot_urls[], created_at, updated_at bug_activity → id, bug_id,
changed_by, field_changed, old_value, new_value, changed_at meetings → id, project_id,
title, type, description, meeting_date, start_time, end_time, timezone, recurrence,
recurrence_end_date, google_calendar_event_id, google_meet_link,
reminder_minutes_before, status, created_by, created_at, updated_at meeting_attendees →
id, meeting_id, contributor_id, rsvp_status milestones → id, project_id, title,
description, target_date, status, achieved_at, announced, created_by, created_at,
updated_at milestone_progress → id, milestone_id, logged_by, progress_note,
progress_percent, blockers, logged_date, created_at essential_sections → id, project_id,
title, description, icon, position, created_by, created_at essential_entries → id,
section_id, project_id, label, data_type, value_text, value_file_url, value_file_name,
is_sensitive, position, note, created_by, created_at, updated_at
```


Email Notification System

All emails use a shared HTML wrapper with the DEVCON+ PM brand, sent from `DEVCON+ PM` `<onboarding@resend.dev>`.

TRIGGER	RECIPIENT	TEMPLATE / CONTENT
Contributor added	New contributor	Welcome email + login CTA
Task assigned	New assignee	Task details + status + due date
Bug assigned	New assignee	Bug severity + description
Meeting scheduled	All attendees	Date, time, timezone, Meet link
Meeting cancelled	All attendees	Cancellation notice
Meeting reminder	All attendees	Starts in N minutes
Milestone achieved	All contributors	Celebration email
Board activity (non-admin)	Admin only	Actor + action + entity + timestamp PHT

Email Reliability: Activity email failures silently swallow errors and return `{ ok: true }` — board operations never fail due to email delivery issues.

Non-Functional Requirements

NFR-1 Performance

Initial page load uses server-side data fetching — the board is populated without client-side waterfalls. Real-time updates (Supabase Realtime) are incremental.

NFR-2 Security

- Admin email is hardcoded server-side (`lib/permissions.ts`), never derived from client input
- Service role key used only in trusted API routes (`/api/notify`), never exposed to the browser
- Soft deletes on contributors — revoked access is immediate but data is preserved

- Sensitive essentials entries are masked by default

NFR-3

Resilience

- Activity email failures silently swallow errors — board operations never fail due to email issues
- Admin bypass prevents permanent lockout if the admin's contributor record is accidentally deleted
- Board mutations are optimistic with rollback on Supabase error

NFR-4

Accessibility

All interactive elements have `title` attributes. Color is not the only differentiator — status also has a text label alongside the color badge.

NFR-5

Responsive Design

Full mobile support: collapsible sidebar drawer (top bar + hamburger) with the same navigation as desktop.

SECTION 10

Out of Scope (Current Version)

- Multi-admin / role-based admin elevation (admin is always a single hardcoded email)
- Public API or webhook integrations beyond Resend and Telegram
- Time tracking per task
- Comments/threads on tasks
- Git integration beyond manual PR link field
- Dark mode
- Exportable reports (CSV/PDF)

SECTION 11

Open Questions & Future Considerations

QUESTION	NOTES
Should guest activity be attributed to a name?	Could add a "Who are you?" one-time prompt stored in localStorage
Should group colors be stored in Supabase?	Current localStorage approach loses colors when clearing browser data; a color column on groups would persist cross-device
Rate limiting on /api/activity?	High-frequency edits (e.g., typing in title) could send many emails; consider debouncing or only emitting on significant field changes
Supabase RLS for open board?	Anon key needs SELECT/INSERT/UPDATE/DELETE policies on all board tables for guest access to function
Telegram bot integration?	Edge Function stubs exist; implementation deferred to Batch 3

SECTION 12

Glossary

TERM	DEFINITION
Admin	The single owner account (rizzperocho@gmail.com) with full system privileges

TERM	DEFINITION
Contributor	A team member added by the admin with access to all pages except Contributors management
Guest	An unauthenticated visitor with open board access
Group	A named section within a project (analogous to a sprint or feature area) that contains tasks
Accent Color	The manually chosen color assigned to a group, controlling its left border and header tint
Activity Feed	The admin's real-time notification bell showing all board CRUD events
Open Board	The access model where the PM board is publicly readable and editable without login
Portal Dropdown	A React dropdown rendered via createPortal at document.body to escape overflow clipping
Soft Delete	Setting deleted_at on a contributor row rather than removing it, preserving audit history
PHT	Philippine Standard Time (Asia/Manila, UTC+8) — used in activity email timestamps